

Centerline Survey & Mission-Preparation Research

Survey methods, ready-made tools, and CSV/DXF mission workflows for the 3WD Marking Rover project

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Scope: centerline/boundary survey methods, off-the-shelf coordinate-capture tools, CSV/DXF mission preparation practice, and how commercial line-marking competitors handle the same pipeline.

Executive Summary

Two proven industry patterns exist for turning a physical site into a robot-drivable path, and they are not mutually exclusive.

- Template / parametric mode — used by the two market leaders in sports-field line marking (Turf Tank, TinyMobileRobots' TinyLineMarker). No CAD import at all: an RTK base station and a few flagged reference points establish the local coordinate frame, and the robot draws standard field layouts from a library of pre-built templates. Fast (minutes), but limited to well-known regular shapes.
- CAD/GIS-import mode — used for irregular geometry: roads, custom field boundaries, and precision-ag guidance. A path or boundary is designed or captured in survey/CAD/GIS software, exported as DXF, CSV, or a shapefile, georeferenced to real-world RTK/GNSS coordinates, and executed directly. TinyMobileRobots' sister product TinySurveyor (road pre-marking) uses exactly this pattern — DXF/CSV loaded via USB onto a GNSS-guided robot.

The 3WD Marking Rover's existing CSV/DXF ingestion pipeline already follows the second, more generalizable pattern. The main gaps relative to industry practice are not in the driving/control stack (which CLAUDE.md shows is already mature and closed) but upstream, in how coordinates are captured and how CSV/DXF files are structured before they reach the pipeline: an explicit CRS-tagging convention, a standard point/feature-code schema, a DXF layer-naming convention, and — optionally — a lightweight template mode for the common repeatable cases. These are addressed section by section below, with a concrete recommendation set in Section 6.

1. How Surveyors Capture Centerline & Boundary Coordinates

1.1 Traditional methods (still in active use)

Total station traversing — the reference method for high-precision, obstructed, or small-area work. It doesn't depend on sky visibility and delivers sub-centimeter angular/distance precision from an occupied known point. Best suited to tight tolerances near structures or under tree canopy.

GNSS RTK (rover + fixed base, or network RTK/VRS) — has largely replaced chain-and-tape and much total-station work for open-sky corridor surveys. A rover tied to a fixed or virtual reference station gets live centimeter-level fixes in real time.

Static GNSS — long dual-frequency occupations, post-processed rather than real time. Reserved for establishing control points where the highest absolute accuracy is needed but real-time isn't.

Leveling (differential/spirit level) — still used for precise vertical control (drainage grades, benchmarks) where GNSS vertical accuracy isn't sufficient.

Chain/tape — survives mainly for small-scale, low-budget, or check measurements.

A documented field workflow for roads: load a feature-code list into the data collector, walk or drive the centerline collecting points coded "CL" (auto-joined into a polyline by the collector), plus offset corridor shots at regular intervals (e.g. every ~25 m) to build a cross-sectional surface, then post-process in office software into a 3D corridor model.

1.2 Modern methods

Mobile mapping systems (MMS) — vehicle-mounted LiDAR + IMU + GNSS (an inertial navigation system) capture dense corridor point clouds while driving at traffic speed. Roughly 1–10 cm accuracy depending on GNSS/IMU quality and processing. Standard for large-scale road-corridor capture without lane closures.

Drone photogrammetry (structure-from-motion) — ~1 cm horizontal / 2–4 cm vertical RMS with good ground control. Favored over open, well-textured terrain; lower equipment cost than LiDAR.

Drone LiDAR — ~1 cm horizontal / 1–3 cm vertical RMS, and outperforms photogrammetry under vegetation, low-contrast surfaces (snow, bare concrete), or at night.

Terrestrial laser scanning (TLS) — very dense, millimeter-to-centimeter accurate point clouds for small, complex, or safety-critical sites, but slow per setup and needs registration across scan stations. A niche fit for corridor centerlines; common for as-built structure detail.

Satellite-imagery digitizing — used mainly for reconnaissance/planning-level boundary estimates given meter-level resolution limits, not construction-grade centerline capture.

1.3 What "centerline" actually means — and how it maps to fields

Three distinct centerlines are easily conflated in road work:

- Design centerline — a CAD/alignment-software geometry (line/curve/spiral stationing) produced during design, independent of ground truth.
- As-built / existing-conditions centerline — captured by a topo survey of what's physically on the ground; used for verification or when no design exists (e.g. re-marking an existing road).
- Stakeout centerline — design coordinates physically re-established on the ground with stakes/marks for construction guidance.

For a marking robot this maps directly. If a design/CAD plan already exists (a new road or a new field layout), the centerline to drive is the design geometry transformed into site coordinates via control points — no field survey needed. If no design exists (re-stripping an existing road, or laying out lines on an existing sports or

agricultural field), the centerline must first be surveyed as-built (walked with RTK/total station, or mobile-mapped), and that captured geometry becomes the reference the robot follows.

Sports-field line-marking systems follow the design path: operators design the pattern in software and transfer it to a GPS-guided vehicle running RTK, which paints to the plan rather than surveying the field itself each time. Agricultural RTK guidance is analogous — rows and boundaries are typically set from a planned AB-line or geofenced boundary rather than surveyed fresh each pass.

1.4 Accuracy, cost, and time tradeoffs

Method	Typical accuracy	Speed / cost	Best fit
RTK GNSS (rover)	1–3 cm horizontal, real time	Fast, low-to-moderate cost	Open-sky corridors and fields
Static GNSS	cm-level, post-processed (hours/occupation)	Slow, moderate cost	Establishing control points
Total station	mm–cm	Slow (one point at a time), needs line of sight	Small or obstructed sites, tight tolerances
Drone photogrammetry	~1 cm horiz. / 2–4 cm vert.	Fast area coverage, lower hardware cost	Open, well-textured terrain
Drone / terrestrial / mobile LiDAR	~1 cm horiz. / 1–3 cm vert.	Higher hardware cost, faster than static TLS for corridors	Vegetation, low light, poor contrast

General rule across sources: GNSS/RTK and photogrammetry are cheapest and fastest for open sites; LiDAR (mobile or drone) costs more but wins under vegetation, obstruction, or poor light; total station/TLS remain the accuracy ceiling for small or occluded areas.

1.5 Standard workflow, site walk to delivered coordinates

- Establish or verify horizontal and vertical control (existing monuments, or new GNSS-tied control points).
- Reconnaissance / site walk to plan the capture method and note obstructions.
- Field data collection (RTK rover traverse, total station traverse, or drone/mobile-LiDAR pass) with coded features (centerline, edges, offsets).
- Download raw data to office software.
- Process and adjust (least-squares traverse adjustment, GNSS baseline processing, or point-cloud registration/classification).
- Build the surface/alignment model and QA against known controls or redundant shots.
- Export final coordinates/alignment in the required format (DXF/CSV, state-plane or local grid) for downstream use — e.g. import into a robot's path-planning stack.

2. Ready-Made Tools for Getting Centerline Coordinates

2.1 GNSS RTK field data collectors

Tool	What it is / notes	Tier
Trimble Access / TSC7	Industry-standard field software for professional crews; tightly integrated with Trimble's own GNSS/robotic total-station hardware. Widely called the “gold standard,” with heavy vendor lock-in.	Premium (\$15–30k+ hardware+software)
Trimble Catalyst	Subscription GNSS positioning on any Android/iOS device via the low-cost Trimble DA2 receiver. Accuracy is selectable, sub-meter to centimeter, per subscription tier (hourly/monthly/annual).	Low-capex entry to Trimble ecosystem
Leica Captivate / Leica ICON	Leica's equivalent premium field software; similar price tier and reputation to Trimble Access.	Premium
Carlson SurvCE / SurvPC	Hardware-agnostic field software running on many GNSS/total-station brands; perpetual license (vs. subscription), optional ~10%/yr maintenance.	Budget/flexible alternative
Emlid Reach RS3 + Flow app	Multi-band GNSS receiver with tilt compensation, ~7mm+1ppm horizontal / 14mm+1ppm vertical RTK. Free Emlid Flow app for field collection/stakeout, Emlid Studio for post-processing, Emlid Caster for NTRIP.	~\$2,999 — realistic budget path for a small team

2.2 Drone photogrammetry / LiDAR mapping software

Pix4Dmapper + Pix4Dsurvey — Pix4Dmapper is an industry-standard, engineering-grade point-cloud/orthomosaic processor; Pix4Dsurvey is the companion vectorization tool that actually produces centerlines and boundaries — it imports point clouds (.las/.laz), generates a TIN, and provides dedicated tools for points, polylines, polygons, and catenary curves to extract curbs, footprints, and breaklines, then exports CAD/GIS-ready files. This is the most direct “point cloud → vector line” ready-made workflow available.

DroneDeploy — cloud-based, strong on orthomosaics/DSMs and volumetric analysis, easy UX; pricing tiers can constrain smaller teams.

Agisoft Metashape — technically strong, but Russian-developed — worth noting if the project ever pursues municipal/government contracts with sourcing restrictions.

WebODM / OpenDroneMap — genuinely free and open source; generates orthomosaics, point clouds, and DEMs, but centerline/vector extraction is not built in to the same degree as Pix4Dsurvey — would need QGIS or Global Mapper downstream for vectorization.

2.3 Desktop CAD/GIS tools

Tool	Notes	Cost
AutoCAD Civil 3D	De facto US industry standard for corridor/centerline design from survey data; deeply embedded in DOT/municipal workflows.	Paid, subscription
Bentley OpenRoads / MicroStation	Civil 3D's main rival, dominant in some infrastructure/rail/DOT markets. Choice is largely about existing client ecosystem.	Paid, comparable to Civil 3D
Carlson Civil	Cheaper alternative aimed at smaller survey/civil firms; perpetual licensing; respected budget-tier tool.	Paid, lower tier
QGIS + Road Centerline Extractor plugin	Free/open source. The plugin converts road polygon layers into centerlines automatically (T-junction segmentation, spur pruning, min-width filtering) — needs a polygon layer as input.	Free

Tool	Notes	Cost
Global Mapper (+ LiDAR Module/Pro)	Mid-priced, well regarded for LiDAR processing (classification, terrain painting); common cheaper alternative to full ArcGIS/Civil 3D stacks.	Paid, mid-tier
ArcGIS Pro	GIS-side industry standard, especially for municipal/agency boundary data; QGIS is functionally competitive for most centerline/digitizing tasks.	Paid, per-seat

2.4 AI / semi-automated centerline extraction — a reality check

Automated extraction tools exist for both LiDAR and imagery, but they are not yet reliable enough to run unsupervised. A 2026 study on LiDAR-based road/centerline extraction reported only 67–73% IoU accuracy even after refinement, explicitly framing results as groundwork requiring human validation before operational use. CNN-based imagery-extraction papers consistently pair automated tracing with manual cleanup. Bottom line: today's “AI-based” centerline tools speed up a first draft but do not replace a manual/semi-manual QA step — plan for a human-in-the-loop review regardless of which product is used.

2.5 Free-path summary

Category	Industry-standard (paid)	Small-team-viable free/OSS path
RTK GNSS	Trimble Access/Catalyst, Leica Captivate	Emlid Reach RS3 + Flow (~\$3k, cm-accuracy)
Drone mapping	Pix4D, DroneDeploy	WebODM / OpenDroneMap
Vectorization	Pix4Dsurvey	QGIS + Road Centerline Extractor plugin
CAD/GIS	Civil 3D, OpenRoads, ArcGIS Pro	QGIS, Global Mapper (low-cost, not free)
Auto-extraction	None fully trusted yet	Same OSS tools + mandatory manual QA pass

3. Preparing CSV / DXF Mission Files — Professional Practice

3.1 CSV conventions

The dominant survey CSV schema is PNEZD — Point ID, Northing, Easting, Elevation, Description — used natively by Carlson, Trimble, and most CAD/survey packages. Column order varies by vendor (PENZD vs. PNEZD vs. “Point, X, Y, Z, Description”), so column-order ambiguity is the single most common real-world import bug; professional tools always force the user to explicitly map columns and confirm units/CRS at import rather than assume order.

Coordinate reference systems: local grid (arbitrary origin, no datum), geographic WGS84 (lat/lon), and projected UTM/State Plane (northing/easting in meters or US survey feet) are fundamentally different and non-interchangeable without an explicit transform. Every serious tool (Civil 3D, GDAL/OGR, QGIS) requires the source and target EPSG codes to be declared, never inferred.

Point/feature codes (“EP” edge of pavement, “CL” centerline, “FH” fire hydrant, etc.) come from vendor feature-code libraries that also carry control codes to auto-draw linework directly from coded field points (start-line/end-line, curve).

3.2 DXF conventions

Civil 3D distinguishes plain polylines (2D, no station/elevation semantics) from 3D polylines (per-vertex elevation, no true horizontal/vertical geometry) from true Alignment objects (parametric tangents/curves/spirals with station-based referencing, superelevation, profiles). Alignments only exist as Civil 3D objects — when exported as plain DXF for non-CAD consumption, they collapse to ordinary polylines/arcs, losing station data, superelevation, and design intent.

Common pitfalls when non-CAD software (a robotics pipeline) consumes DXF:

- No embedded CRS or units — DXF has no mandatory geodetic reference, so georeferencing must be done manually against known control points.
- Layer-naming conventions (e.g. “C-ROAD-CNTR” style) are the only reliable way to filter the right entities.
- Arcs and splines are frequently exploded or coarsely faceted into line segments by exporters, silently degrading curve fidelity.
- Blocks used for point markers carry attribute data (point ID, code) that many DXF parsers ignore unless explicitly walked.

QGIS plugins such as Save-DXF and AnotherDXFImporter are the standard free route for georeferencing/reprojecting DXF into a real CRS before use.

3.3 “Mission” formats in robotics and precision ag

MAVLink / PX4 / ArduPilot — the legacy plain-text .waypoints/QGC WPL format, and the modern QGC .plan JSON format (mission items + optional GeoFence polygons + RallyPoints, each carrying a MAV_FRAME and altitude mode). PX4’s ROS2 API also exposes waypoint missions programmatically.

ROS2 Nav2 — nav2_waypoint_follower and a dedicated GPS Waypoint Follower consume a list of GPS or map-frame poses and drive Nav2’s planner/controller between them, rather than a single fixed file format — the “mission” is typically assembled at runtime from a script or config listing lat/lon/yaw goals.

Precision ag — ISO 11783 ISOXML (TASKDATA.XML) is the vendor-neutral standard exchanged between farm-management software and in-cab task controllers. It encodes fields as Partfields with boundary polygons, guidance lines/patterns, and task/product data, and is bidirectional (plan out, as-applied data back). AgOpenGPS (open source) implements a comparable guidance-line/boundary model outside ISOXML.

In practice, conversion from surveyor CSV/DXF into any of these mission formats is almost always a custom script (Python/GDAL/pyproj plus a mission-format writer), not an off-the-shelf converter — this matches the pattern the rover’s own pipeline already follows.

3.4 Purpose-built mission-planning tools

Tool	Notes	Free / paid
QGroundControl	Plan View lets you draw or import polygons and generate survey-grid missions; exports .plan.	Free / open source
Mission Planner (ArduPilot)	Mature waypoint editor with CSV import support.	Free / open source
QGIS	De facto standard for georeferencing, CRS handling, and DXF ingestion (not path planning itself).	Free / open source
ROS2 Nav2	Planning/execution framework, not file authoring.	Free / open source
Trimble Ag / Precision-IQ	Farm-data compatibility, ISOXML import/export.	Paid, commercial

Tool	Notes	Free / paid
John Deere Operations Center, Ag Leader SMS	Robust, widely deployed ISOXML-centric guidance/task systems.	Paid, commercial

For the marking-robot use case specifically, TinyMobileRobots' TinySurveyor (road pre-marking robot) is the most directly comparable commercial product: it has marked over a million fields, and paths are designed off-robot in CAD/GIS software, exported as DXF or CSV, and loaded via USB onto a GNSS-guided robot compatible with Trimble/Leica/Topcon receivers — a strong signal that this CAD-to-drivable-path pipeline is production-hardened.

3.5 Best-practice recommendations from the research (schema level)

- Pin one canonical internal CSV schema — a PNEZD-superset: point_id, northing_or_lat, easting_or_lon, elevation_m, heading_deg (optional), code, crs_epsg — and reject or flag any import missing the CRS field. Never assume it.
- Do all CRS transforms with pyproj/GDAL against explicit EPSG codes at import time, with a logged before/after sanity check (point count, bounding box). Silent CRS/units mismatch is the single biggest field-failure mode across every source reviewed.
- For DXF ingestion, standardize a required layer-naming convention (e.g. MARK-CENTERLINE, MARK-BOUNDARY) and always run the file through a georeferencing pass (QGIS or ezdxflib) before consuming — never trust implicit scene units.
- Keep the mission format ROS2-native (a Nav2-style pose list, or the project's own PositionTarget-driven mission JSON) rather than shoehorning into .plan/QGC WPL — those are aviation-specific frames (MAV_FRAME, relative-alt) that don't map cleanly to ground-vehicle OFFBOARD velocity control. This matches what the project already does.
- Given the spray/marking use case, ISOXML's guidance-line + boundary model (segment/boundary separation, product on/off gating) is the closest industry analog worth studying for schema ideas, even without adopting the XML format itself.

4. How Commercial Line-Marking Competitors Do It

4.1 Turf Tank

Turf Tank does not use CAD import or manual “teach by driving.” The workflow is: place an RTK-GNSS base station at the field, flag the corners, then use a tablet drag-and-drop app with a library of pre-built sport-specific line layouts. The base station and cloud-synced RTK establish the coordinate frame; the app generates the executable line geometry parametrically from the corner points and template rules — not from a surveyed point cloud. An offline mode exists for poor-connectivity sites.

Market position: strong incumbent (founded 2015), 1,200+ robots deployed in the US / 1,500+ globally, customers including major US universities, two NFL stadiums, and MLS teams. Reduces marking time from hours to roughly 30 minutes (soccer/baseball) or 3–4 hours (football, vs. 8–10 manual). Pricing: purchase ~\$43k, or subscription \$6k–\$16k/yr depending on tier, first-year all-in \$7.5k–\$19.5k.

4.2 TinyMobileRobots (TinyLineMarker and TinySurveyor)

TinyLineMarker Pro X / Sport follows the same no-manual-survey philosophy as Turf Tank: 200+ pre-built templates covering 50+ sports, selected on a rugged tablet and tweaked for dimensions or logos. Positioning

uses multi-constellation GNSS (GPS/Galileo/GLONASS/BeiDou) plus network RTK, claiming 1–2 cm accuracy and roughly 5-minute field setup; layouts are cloud-stored and reusable across venues. Market position: the direct #2 competitor to Turf Tank, broad dealer network, frequently compared head-to-head in industry coverage.

The company's sister product, TinySurveyor, is aimed at road pre-marking/stripping and is the closest analog to a CAD-driven pipeline in this space: paths are designed off-robot in CAD/GIS software, exported as DXF or CSV, and loaded via USB onto the robot, which executes autonomously using GNSS compatible with Trimble/Leica/Topcon receivers — materially different from, and more “surveying-industry-standard” than, the sports-field template approach.

4.3 Other line-marking robots

Other GPS-guided semi-autonomous striping machines aimed at DOT/paving contractors include RoboTRAFFIC RTS (USSC), the Tyker Robot Plotter (BauMotor / C.R. Kennedy), SWOZI auto, and Traqnology. These are smaller and less mainstream than Turf Tank or TinyMobileRobots, positioned as productivity tools for existing striping crews rather than fully autonomous fleets, with pricing generally quote-based rather than public.

4.4 Adjacent precision agriculture and robotic mowing

Carbon Robotics' LaserWeeder/autonomous tractor kit uses GPS plus computer-vision furrow-following with automatic headland turns, but the boundary-to-path algorithm details and file formats are proprietary and not publicly documented.

John Deere Operations Center / AutoPath documents a clear, publicly disclosed pattern: boundaries are captured (driven once, or imported as shapefiles from Operations Center, Pix4Dfields, fieldmargin, GeoPard, or similar), and the platform then auto-generates parallel guidance passes (AB lines) from that boundary plus row spacing and implement width. This “record-boundary-once, then auto-generate parallel coverage” pattern is the de facto industry standard across precision ag, and is echoed in coverage-path-planning research for robotic/commercial mowers: teach the perimeter (drive it once, or draw it), then run a boustrophedon (back-and-forth) or cellular-decomposition coverage planner to fill the interior automatically.

4.5 The two converging patterns, and what a small team can realistically replicate

Template / parametric mode (Turf Tank, TinyMobileRobots' TinyLineMarker): skip surveying entirely; use a fixed real-world reference (RTK base station + flagged corners, or a single teach point) and generate geometry from known standard field dimensions. Fast, no CAD pipeline needed, but limited to well-known, regular shapes.

CAD/GIS-import mode (TinySurveyor, ag guidance platforms): design the path or boundary in existing GIS/CAD tools, export as DXF, shapefile, or CSV waypoint list, georeference it to RTK/GNSS coordinates, and execute directly. This is the more generalizable, industry-standard pattern for irregular geometry (roads, custom fields), and the one most transferable to a DXF-driven pipeline that already exists.

A small team without a mapping department can realistically combine both: keep robust DXF/CSV import with a simple CRS/geodesic converter for irregular jobs, and add a lightweight template/base-station mode for common repeatable shapes to cut field time for the common case — mirroring exactly how Turf Tank and TinyMobileRobots minimize setup. Full boundary-driven coverage planning (drive-once-then-auto-fill) is worth deferring unless the customer base needs full-area coverage (mowing/ag-style) rather than discrete line/curve marking.

5. Recommendations for the 3WD Marking Rover Project

These recommendations connect the research above to the project's current stage — controller and OFFBOARD tracking are closed and validated at the trusted baseline; the open work is path engine/trajectory planning, CRS/geodesic handling for path import, and full-pipeline validation from CAD/DXF through to drive and spray.

- Formalize the CSV schema now, before more mission data accumulates: point_id, lat/lon or northing/easting, elevation (optional), heading (optional), code, and a mandatory crs_epsg field. Reject any file missing the CRS tag rather than guessing it.
- Add an explicit CRS-transform step using pyproj/GDAL with a logged sanity check (bounding box, point count) immediately after CSV/DXF ingestion — this is the single highest-leverage fix given how consistently CRS/units mismatches showed up as the top real-world failure mode across every source reviewed.
- Adopt a layer-naming convention for incoming DXF files (e.g. MARK-CENTERLINE, MARK-BOUNDARY) and run every DXF through a georeferencing pass before the path engine touches it, since DXF carries no mandatory CRS or units.
- For a low-cost path to real survey-grade coordinates when no design CAD exists (re-marking an existing field or road), an Emlid Reach RS3 + Flow setup (~\$3k, cm-level RTK) is a realistic small-team option — far below Trimble/Leica field-collector pricing — for building the as-built CSV directly.
- Consider a template/base-station mode for the most common, regular field layouts (mirroring Turf Tank/TinyLineMarker), reserving the full DXF/CSV import path for irregular geometry — this could meaningfully cut setup time for repeat customers without adding CAD-pipeline risk.
- Treat any future AI/automatic centerline-extraction tooling as a first-draft assist only; published accuracy (67–73% IoU even after refinement) is not yet reliable enough to skip a human QA pass.
- Do not adopt QGC .plan / MAVLink waypoint files as the mission format — they encode aviation-specific frames that don't map cleanly onto ground-vehicle OFFBOARD velocity control. A Nav2-style pose list or the project's own PositionTarget-driven JSON remains the right fit, matching current practice.
- When designing the CSV/DXF schema's segment and boundary concepts, use ISOXML's guidance-line/boundary separation and product on/off gating as a reference model — it is the closest publicly documented industry analog to a line/spray mission format, even without adopting the XML format itself.

Sources

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